

EcoSphere ESG Management Platform

Team Architecture Blueprint

Project Vision

EcoSphere is an enterprise ESG platform that tracks Environmental, Social and Governance activities inside a company. It does not perform business operations; instead, it records operational activities, converts them into ESG metrics, gamifies employee participation, and provides management dashboards and reports.

Core Workflow

1. Admin configures the company (departments, employees, emission factors, policies, rewards, badges).
2. Department Managers perform daily operations by entering operational data, creating CSR activities/challenges and approving employee submissions.
3. Employees participate in CSR/challenges, upload proof, accept policies, earn XP and redeem rewards.
4. Auditors inspect the company, raise compliance issues and verify resolutions.
5. The system automatically updates ESG scores, leaderboards, notifications and analytics.

Roles

1. Admin (CEO + ESG Manager)

- Executive dashboard with ESG analytics and reports.
- Department overview and drill-down.
- Manage departments, employees, emission factors, policies, rewards, badges and categories.
- Export reports (PDF/CSV/Excel).

2. Department Manager

- Department dashboard.
- Operations (mini ERP): purchase, electricity, fleet, manufacturing, waste entries with bill/image/video/PDF proof.
- Create CSR activities and challenges.
- Approve employee submissions.
- Monitor employees, policies, compliance issues and department analytics.

3. Employee

- Dashboard showing XP, level, rewards, badges and rankings.
- Join CSR activities and challenges.
- Upload proof.
- Accept company policies.
- Redeem rewards.
- View notifications and leaderboards.

4. Auditor

- Perform audits.
- Raise compliance issues.
- Assign owners and deadlines.
- Track issue history.

Mini ERP (Operations)

Instead of integrating SAP/Odoo, the application contains an Operations section where department managers enter daily operational data. Examples include diesel purchases, electricity usage, fleet

fuel, manufacturing logs and waste generation. The system converts these entries into carbon transactions automatically.

Environmental Module

Operational data → Carbon calculation using emission factors → Department Environmental Score
→ Organization ESG Dashboard.

Social Module

Managers create CSR activities and sustainability challenges. Employees participate, upload proof and receive XP after approval. Rewards and badges are assigned automatically based on predefined rules.

Governance Module

Admin publishes company policies. Employees read and accept them. Auditors verify compliance, raise issues when violations are found, and managers resolve those issues.

Gamification

- XP earned from CSR, challenges and trainings.
- Badges automatically assigned when rules are satisfied.
- Rewards redeemed using earned points/coins.
- Department and company leaderboards.

Notifications

In-app and email notifications for:

- New policy
- Challenge approval
- Badge unlocked
- Reward available
- Compliance issue assigned
- Reminders and deadlines

Final Navigation

Admin: Dashboard, Departments, Rewards & Badges, Master Data, Settings.

Manager: Dashboard, Operations, CSR & Challenges, Employees, Policies, Compliance, Analytics.

Employee: Dashboard, Challenges, CSR, Rewards & Badges, Leaderboard, Policies, Notifications.

Auditor: Dashboard, Audits, Compliance Issues, History.

Guiding Principle

People perform normal work. EcoSphere records that work, converts it into ESG insights, rewards participation and provides decision-making dashboards.